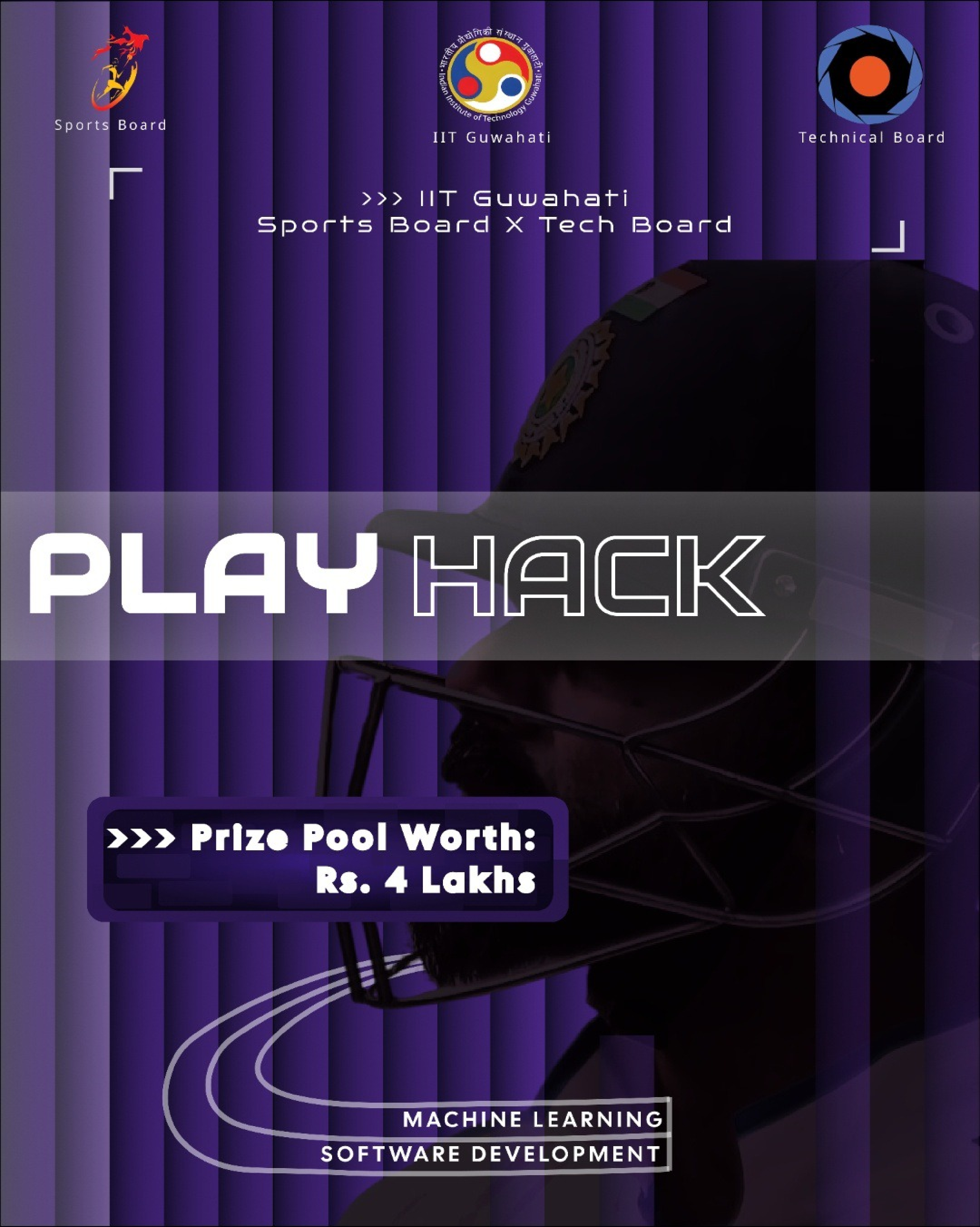
Sports Board
IIT Guwahati
Technical Board
>>> IIT Guwahati
Sports Board X Tech Board
PLAY HACK
>>> Prize Pool Worth:
Rs. 4 Lakhs
MACHINE LEARNING
SOFTWARE DEVELOPMENT

PLAYHACK / ML PS

Will this athlete get injured next month, and for how long?

THE CHALLENGE

PLAYHACK / ML PS looks at a month of training data and predicts whether an injury will occur during the next month. If an injury is expected, the model must also estimate when it will begin and how long it will sideline the athlete.

One athlete. Two windows. Three predictions.

OBSERVATION WINDOW

Days 1 to 30 are provided in full. Use only this window to build your features and predictions.

RISK WINDOW

Days 31 to 60 are hidden. Predict what happens in this future 30-day period.

TASK A

Binary injury classification: does an injury onset occur during the risk window?

TASK B

For predicted injuries, estimate onset day offset and recovery duration.

Make the future measurable.

Every submission must describe both the event and its timing.

A

`injured_in_risk_window`

Predict 0 or 1. A value of 1 means an injury onset occurs during days 31 to 60; 0 means it does not.

B

`onset_day_offset`

For athletes predicted injured, predict an integer from 1 to 30 indicating which day of the risk window the injury starts.

C

`recovery_duration`

For athletes predicted injured, predict the number of recovery days that follow the injury onset.

IMPORTANT SUBMISSION RULE

`onset_day_offset` and `recovery_duration` are required for every athlete in `sample_submission.csv`, even when `injured_in_risk_window` is predicted as 0.

Use only the observation window for modeling. The risk window and its labels remain hidden for test athletes.

Many signals. One athlete story.

Join daily, hourly, sleep, body, training, and metadata signals by athlete.

ACTIVITY AND BIOMETRICS

dailyActivity_merged.csv	Id, ActivityDate, TotalSteps, TotalDistance, TrackerDistance, VeryActiveMinutes, FairlyActiveMinutes, LightlyActiveMinutes, SedentaryMinutes, Calories
hourlySteps_merged.csv	Id, ActivityHour, StepTotal
hourlyCalories_merged.csv	Id, ActivityHour, Calories
hourlyIntensities_merged.csv	Id, ActivityHour, TotalIntensity, AverageIntensity
hourlyHeartrate_merged.csv	Id, ActivityHour, AvgHeartRate, MinHeartRate, MaxHeartRate

RECOVERY CONTEXT

sleepDay_merged.csv	Id, SleepDay, TotalSleepRecords, TotalMinutesAsleep, TotalTimeInBed
weightLogInfo_merged.csv	Id, Date, WeightKg, Fat, BMI, IsManualReport

Know your data. Build your model.

The strongest solutions connect domain signals to a disciplined evaluation strategy.

ADDITIONAL FILES

training_sessions.csv	session_id, athlete_id, date, start_hour, end_hour, sport_session_type (practice / scrimmage / gym)
athlete_metadata.csv	athlete_id, sport, age, gender, height_cm, weight_kg_baseline, dominant_side, years_playing, position, team_id, prior_season_injury_count
train_labels.csv	athlete_id, injured_in_risk_window, onset_day_offset, recovery_duration

SUBMISSION FILE

sample_submission.csv must contain athlete_id, injured_in_risk_window, onset_day_offset, and recovery_duration. The two regression predictions are required even when the classification prediction is 0.

MODELING REMINDER

Build features from the observation window only. Keep the classification decision and the two timing predictions consistent with one another.

Evaluation metrics

Consistent measures for classification, timing accuracy, and model skill.

Task A: F1 score

F1-score, in [0, 1]. Measures whether an injury onset occurs during the risk window.

Task B: timing

Evaluated on every test athlete truly injured in the risk window. Score onset_day_offset and recovery_duration.

Hit: injured

Mean absolute error (MAE) is calculated against the true onset_day_offset and recovery_duration.

Miss: not injured

A fixed penalty of PENALTY = n_risk = 30 applies to both timing predictions when an injury is missed.

Skill score

```
skill = max(0, 1 - MAE_model / MAE_baseline)
```

Each MAE becomes a skill score on this scale.

Baseline

Predicts the training-set mean onset day and recovery duration for every injured athlete.

Build PLAYHACK / ML PS.

Find the signal before the setback. Turn athlete data into earlier, more useful decisions.